

User Interface of the Web Application

STeMP
v0.1

What is
STeMP?

How to use
this app

Upload data

Create a
protocol

Protocol
Viewer



Upload model (.RDS)

Upload model object (.RDS)

Browse... model.RDS

Model loaded successfully

Delete uploaded model

Upload geospatial data (.gpkg)

Upload sampling locations

Browse... tpoints.gpkg

Delete uploaded samples

Upload prediction area

Browse... modeldomain.gpkg

Delete uploaded prediction area

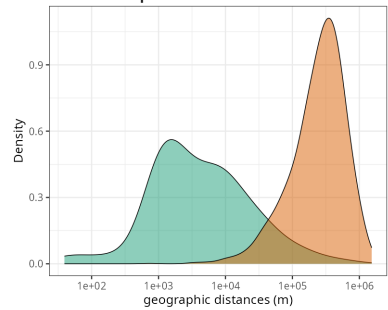
Section: Prediction

Prediction domain

Prediction map



Geodistance plot



distance function sample-to-sample prediction-to-sample

Section: Model

Random resampling might yield overly optimistic results with clustered samples. See Roberts et al., 2017, Milà et al., 2022

Model validation and selection

Validation strategy *

Random Cross-Validation

Performance metrics *

RMSE Rsquared

Validation results *

RMSE = 22.330, $R^2 = 0.737$

Hyperparameter tuning *

mtry=6, splitrule=1, min.node.size=5

Create a report from the completed Protocol

```
library(STeMP)
protocol <- read.csv("protocol.csv")
analyze_protocol(protocol)
```

Warnings

- 1 Cross-Validation was used for both model selection and evaluating the final prediction. This can lead to data leakage. See [Hastie et al., 2009](#).
- 2 Random Cross-Validation was used for model selection, while the training samples were clustered relative to the prediction area. This might yield overly optimistic results and thus bias model selection, or false confidence in the models' performance if no final map accuracy is estimated. See [Wadoux et al., 2021](#), [de Bruin et al., 2022](#), [Milà et al., 2022](#).
- 3 Random Cross-Validation was used for final map accuracy estimation, while the training samples were clustered relative to the prediction area. This might yield overly optimistic results and give false confidence in the resulting map. See [Wadoux et al., 2021](#), [de Bruin et al., 2022](#), [Milà et al., 2022](#).
- 4 Using spatial proxies with clustered samples likely leads to extrapolation situations. See [Meyer et al., 2019](#), [Milà et al., 2024](#).
- 5 The samples were clustered relative to the prediction area, which often leads to extrapolation when the model is applied to feature combinations not present in the training data. No methods to identify and communicate extrapolation areas were reported. See [Meyer & Pebesma, 2021](#), [Elith et al., 2010](#).